Experimental Probability and Expected Outcomes

Name _____ Date _____ Calculator allowed Version 1

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KEY WORDS experiment *(whakamātau)* · trial *(whakamātautau)* · frequency *(auautanga)*

A Relative frequency ●○○

Remember: Relative frequency (experimental probability) = the number of times the event happened ÷ the total number of trials.

1. A spinner with three colours is spun many times. The table shows the results.

Colour	Red	Blue	Green
Frequency	20	12	8

a) How many trials were there altogether? _____

b) Write the relative frequency of red as a fraction in its simplest form. ___

c) Write the relative frequency of blue as a decimal. _____

d) Write the relative frequency of red or green as a fraction. ___

2. A bag holds counters of four colours. A counter is picked, its colour is noted and it is put back. The table shows the results.

Colour	Black	White	Pink	Gold
Frequency	15	20	10	5

a) How many trials were there altogether? _____

b) Write the relative frequency of white as a fraction in its simplest form. ___

c) Write the relative frequency of gold as a decimal. ___

d) Write the relative frequency of not picking black as a fraction. ___

B Estimating from experiments ●●○

Remember: Use the results as an estimate of the probability. The more trials there are, the better the estimate.

3. A drawing pin is dropped 50 times. It lands point up 18 times.

a) Estimate the probability that it lands point up.

b) How many times would you expect it to land point up in 200 drops?

4. In a survey of 80 students, 52 had a pet.

a) Estimate the probability that a student chosen at random has a pet.

b) How many of a school of 600 students would you expect to have a pet?

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C Expected number of outcomes ●●○

Remember: Expected number = probability × number of trials.

5. Work out how many times you would expect each event to happen.

a) A fair coin is tossed 80 times. How many heads would you expect?

b) A fair dice is rolled 60 times. How many fives would you expect?

c) A fair dice is rolled 90 times. How many even numbers would you expect?

d) A spinner has 4 equal sections and one is red. It is spun 100 times. How many reds would you expect?

e) A bag holds 2 red and 3 blue counters. A counter is picked and put back 50 times. How many reds would you expect?

D Experiment and theory ●●●

Watch out: Results from an experiment are never exactly what theory predicts. Look for results that are a long way from what is expected.

6. A dice is rolled 60 times. The table shows the results.

Number	1	2	3	4	5	6
Frequency	11	9	10	12	8	10

a) Find the experimental probability of rolling a 6.

b) Write down the theoretical probability of rolling a 6 on a fair dice.

c) How many times would you expect each number in 60 rolls of a fair dice?

d) Do the results suggest that this dice is fair? Give a reason.

7. A dice is rolled 60 times. The table shows the results.

Number	1	2	3	4	5	6
Frequency	5	6	4	5	6	34

a) Find the experimental probability of rolling a 6.

b) Write down the theoretical probability of rolling a 6 on a fair dice.

c) How many times would you expect each number in 60 rolls of a fair dice?

d) Do the results suggest that this dice is fair? Give a reason.

★ Extension

8. A bag holds 10 counters. A counter is picked and put back 50 times, and it is red 15 times. Estimate the number of red counters in the bag.

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Experimental Probability and Expected Outcomes

Name _____ Date _____ Calculator allowed Version 2

Scan to practise this topic online

KEY WORDS experiment *(whakamātau)* · trial *(whakamātautau)* · frequency *(auautanga)*

A Relative frequency ●○○

Remember: Relative frequency (experimental probability) = the number of times the event happened ÷ the total number of trials.

1. A spinner with three colours is spun many times. The table shows the results.

Colour	Red	Blue	Green
Frequency	25	15	10

a) How many trials were there altogether? _____

b) Write the relative frequency of red as a fraction in its simplest form. ___

c) Write the relative frequency of blue as a decimal. _____

d) Write the relative frequency of red or green as a fraction. ___

2. A bag holds counters of four colours. A counter is picked, its colour is noted and it is put back. The table shows the results.

Colour	Black	White	Pink	Gold
Frequency	12	16	8	4

a) How many trials were there altogether? _____

b) Write the relative frequency of white as a fraction in its simplest form. ___

c) Write the relative frequency of gold as a decimal. ___

d) Write the relative frequency of not picking black as a fraction. ___

B Estimating from experiments ●●○

Remember: Use the results as an estimate of the probability. The more trials there are, the better the estimate.

3. A drawing pin is dropped 50 times. It lands point up 22 times.

a) Estimate the probability that it lands point up.

b) How many times would you expect it to land point up in 200 drops?

4. In a survey of 80 students, 44 had a pet.

a) Estimate the probability that a student chosen at random has a pet.

b) How many of a school of 600 students would you expect to have a pet?

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C Expected number of outcomes ●●○

Remember: Expected number = probability × number of trials.

5. Work out how many times you would expect each event to happen.

a) A fair coin is tossed 120 times. How many tails would you expect?

b) A fair dice is rolled 90 times. How many twos would you expect?

c) A fair dice is rolled 70 times. How many odd numbers would you expect?

d) A spinner has 5 equal sections and one is red. It is spun 100 times. How many reds would you expect?

e) A bag holds 3 red and 2 blue counters. A counter is picked and put back 50 times. How many reds would you expect?

D Experiment and theory ●●●

Watch out: Results from an experiment are never exactly what theory predicts. Look for results that are a long way from what is expected.

6. A dice is rolled 60 times. The table shows the results.

Number	1	2	3	4	5	6
Frequency	10	12	9	11	10	8

a) Find the experimental probability of rolling a 6.

b) Write down the theoretical probability of rolling a 6 on a fair dice.

c) How many times would you expect each number in 60 rolls of a fair dice?

d) Do the results suggest that this dice is fair? Give a reason.

7. A dice is rolled 60 times. The table shows the results.

Number	1	2	3	4	5	6
Frequency	4	5	33	6	5	7

a) Find the experimental probability of rolling a 6.

b) Write down the theoretical probability of rolling a 6 on a fair dice.

c) How many times would you expect each number in 60 rolls of a fair dice?

d) Do the results suggest that this dice is fair? Give a reason.

★ Extension

8. A bag holds 20 counters. A counter is picked and put back 80 times, and it is red 28 times. Estimate the number of red counters in the bag.

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